

NR1 (LVQLL) invariant across 46 species; PGQLP only in monotremes

(a)

NR box 1

Outgroups

<i>L. chalumnae</i>	coelacanth	L	V	Q	L	L
<i>G. seraphini</i>	caecilian	L	V	Q	L	L
<i>B. bufo</i>	toad	L	V	Q	L	L
<i>X. tropicalis</i>	frog	L	V	Q	L	L

Reptiles/birds

<i>C. mydas</i>	turtle	L	V	Q	L	L
<i>G. gallus</i>	chicken	L	V	Q	L	L
<i>C. porosus</i>	crocodile	L	V	Q	L	L
<i>A. carolinensis</i>	anole	L	V	Q	L	L
<i>P. h. viviparus</i> ancestral NR1 PGQLP (PP ≥ 0.999)	python	L	V	Q	L	L

Monotremes

<i>O. anatinus</i>	platypus	P	G	Q	L	P
<i>T. aculeatus</i>	echidna	P	G	Q	L	P

Marsupials

<i>M. domestica</i>	opossum	L	V	Q	L	L
<i>S. harrisii</i>	devil	L	V	Q	L	L

Eutherians

<i>H. sapiens</i>	human	L	V	Q	L	L
<i>M. musculus</i>	mouse	L	V	Q	L	L
<i>B. taurus</i>	cow	L	V	Q	L	L

(b)

Only NR1 changed; NR2/NR3 intact in monotremes

	NR1					NR2					NR3				
Therian (human)	L	V	Q	L	L	L	H	R	L	L	L	R	Y	L	L
Platypus	P	G	Q	L	P	L	H	R	L	L	L	R	Y	L	L
Echidna	P	G	Q	L	P	L	H	R	L	L	L	R	Y	L	L